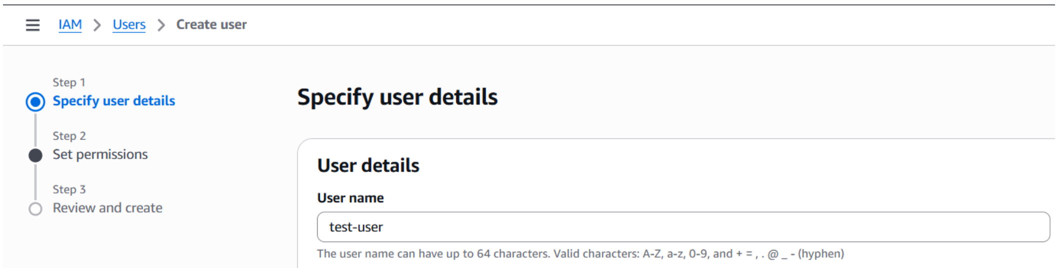
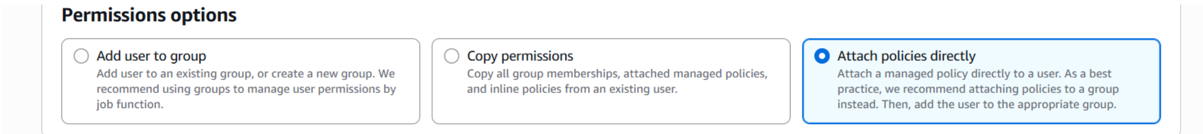
TASK ON IAM

1.Create one IAM user and assign EC2 and S3 full access roles.

- **Logged into the AWS Management Console and navigated to IAM service.**
- **Created a new IAM user named test-user.**

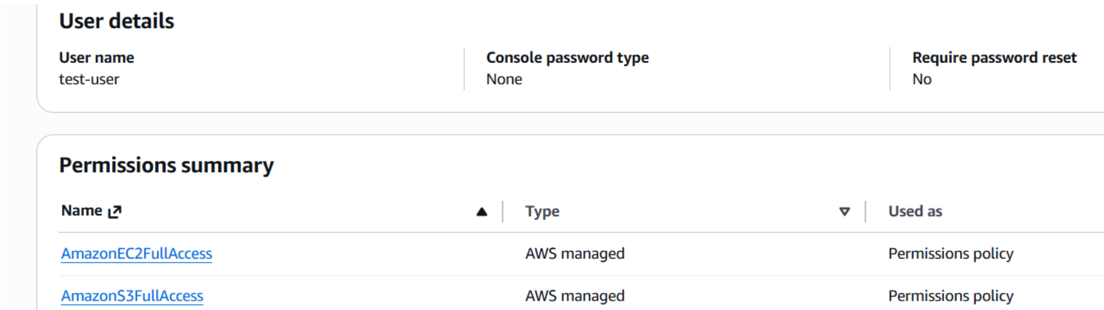


- **Selected Attach policies directly option for assigning permissions.**

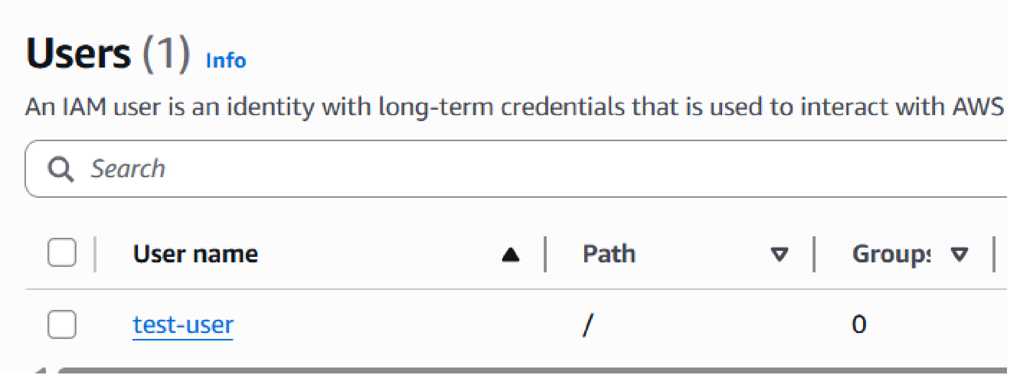


Attached AWS managed policies:

- AmazonEC2FullAccess
- AmazonS3FullAccess

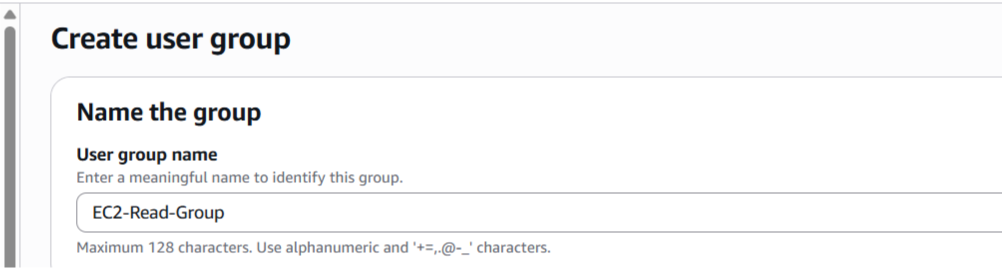


- **Verified permissions under Permissions Summary section.**
- **Successfully created the user with full access to EC2 and S3 services.**

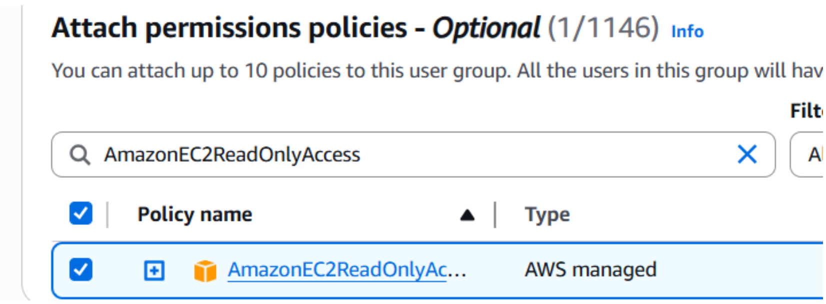


2. Create one group in IAM and assign read access for EC2.

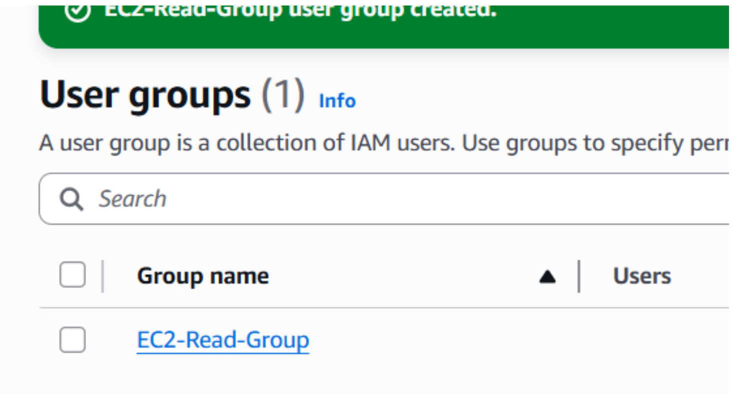
- **Logged into the AWS Management Console and opened IAM service.**
- **Navigated to User Groups and selected Create group.**



- Entered the group name as **EC2-Read-Group**
- In the permissions section, searched for **AmazonEC2ReadOnlyAccess** policy.

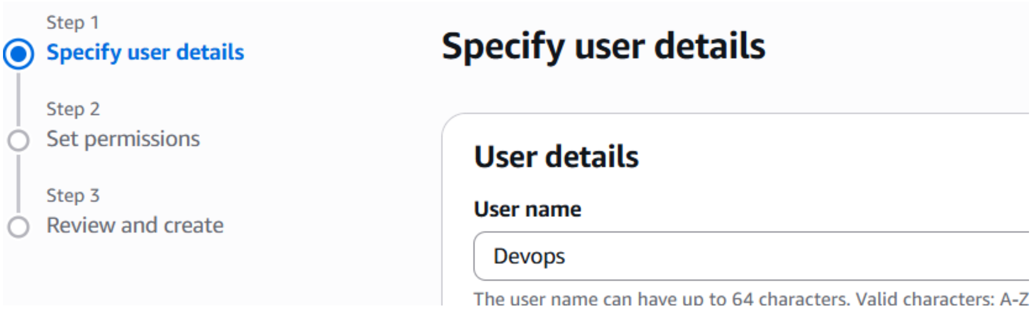


- Completed the group creation process successfully.

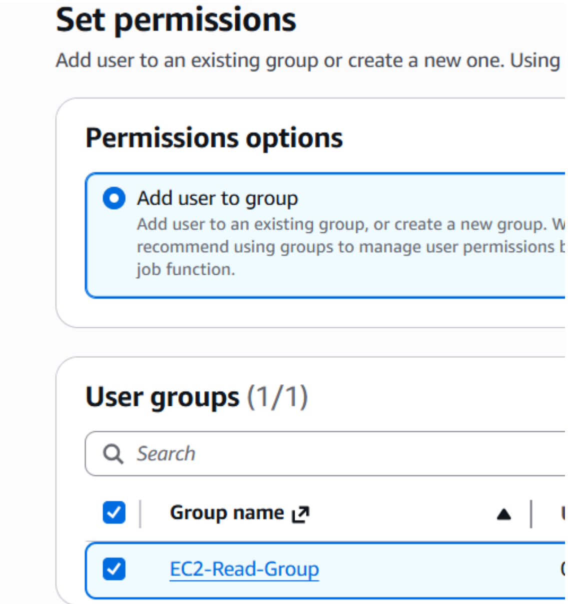


3. **Create a new user named "Devops" and add to the group created in task 2.**

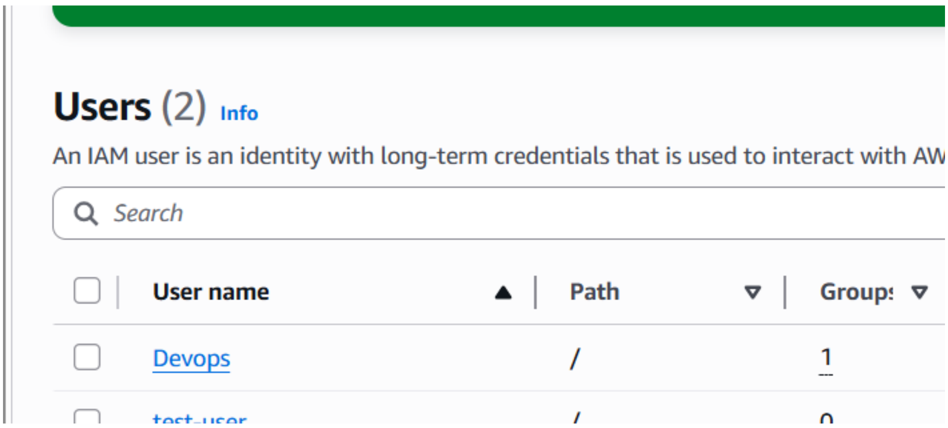
- **Logged into the AWS Management Console and navigated to IAM service.**
- **Selected Users → Create user option.**
- **Entered the user's name as Devops under user details section.**



• Proceeded to **Set permissions** step and **Add user to group** option as **EC2-Read-Group**



- **New user Devops created successfully**



4. Write a bash script to create an IAM user with VPC full access.

- **Installed and configured AWS CLI using aws configure with valid credentials.**

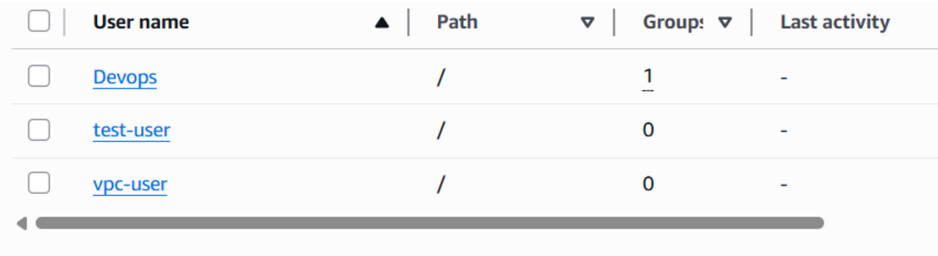
```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ aws --version
aws-cli/2.34.30 Python/3.14.4 Windows/11 exe/AMD64

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$
```

- **Created a bash script file (create-user.sh) to automate IAM user creation.**
- **Executed the script using: ./create-user.sh**

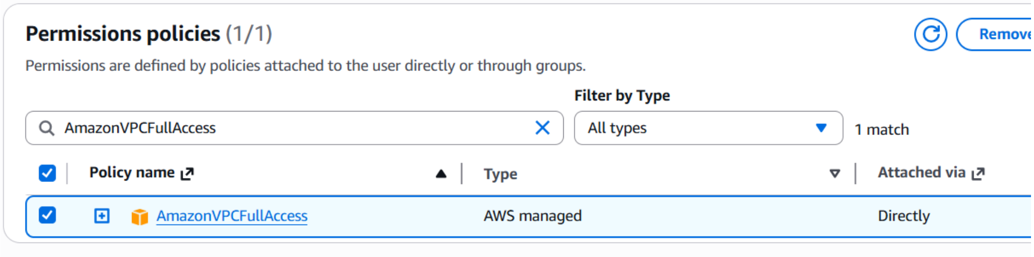
```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ aws configure
AWS Access Key ID [****************VROZ]: AKIA5QKV7AB3Q62GFDNU
AWS Secret Access Key [****************zhzj]: 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
Default region name [us-east-1]: us-east-1
Default output format [json]: json

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ nano create-user.sh

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ ./create-user.sh
{
    "User": {
        "Path": "/",
        "UserName": "vpc-user",
        "UserId": "AIDA5QKV7AB3U65WO42WC",
        "Arn": "arn:aws:iam::928430096503:user/vpc-user",
        "CreateDate": "2026-04-20T11:01:41+00:00"
    }
}
User created successfully

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
```

```
  GNU nano 6.3                  create-user.sh                  Modified
#!/bin/bash

USERNAME="vpc-user"

# Create IAM user
aws iam create-user --user-name $USERNAME

# Attach VPC Full Access policy
aws iam attach-user-policy \
  --user-name $USERNAME \
  --policy-arn arn:aws:iam::aws:policy/AmazonVPCFullAccess

echo "User created successfully"
```

- Defined the IAM user name as **vpc-user** inside the script.
- Used AWS CLI command aws Iam create-user to create the IAM user.



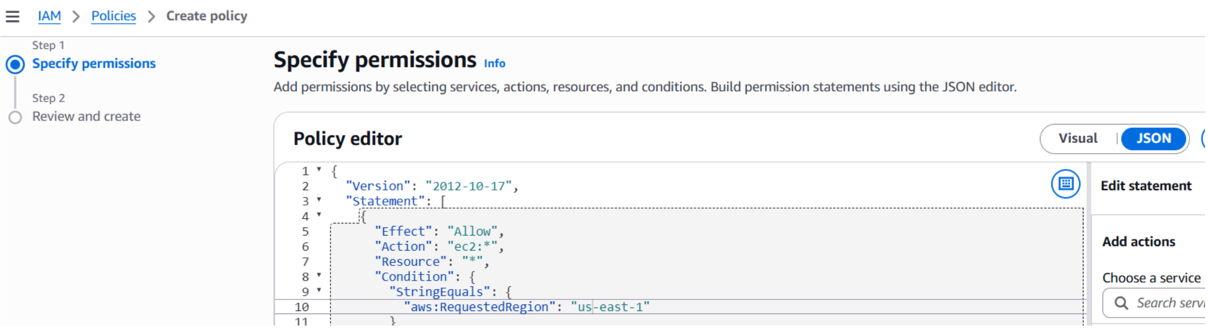
- **Attached AWS managed policy: AmazonVPCFullAccess using aws iam attach-user-policy**



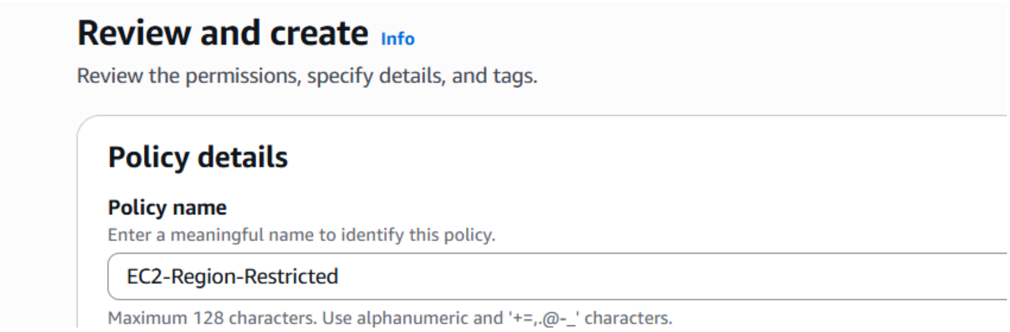
- Confirmed that **AmazonVPCFullAccess** policy is attached to the user.

5. Create an IAM policy to allow EC2 access for a specific user in specific regions only.

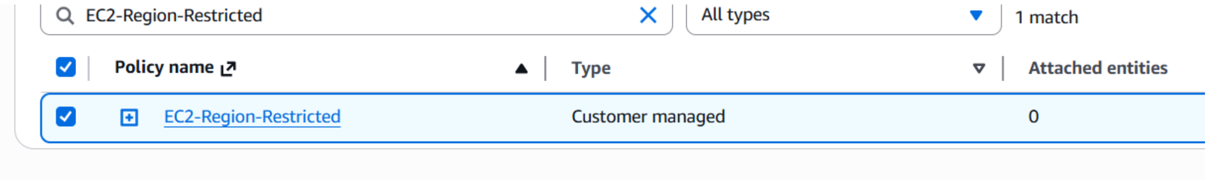
- **Logged into the AWS Management Console and opened IAM service.**
- **Navigated to Policies → Create Policy.**
- **Selected JSON tab to define custom policy.**



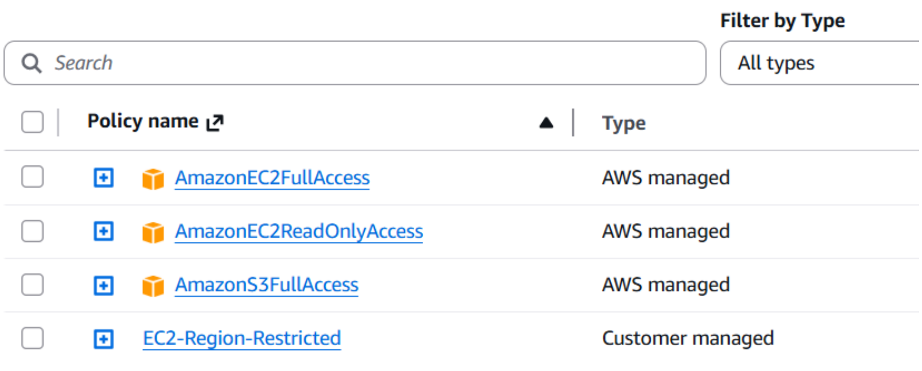
- **Created a policy allowing EC2 actions using:**
- **`"Action": "ec2:*"` and `"Resource": "*"`**
- **Added a Condition block to restrict access to a specific region.**
- **Used condition key: `"aws:RequestedRegion": "us-east-1"`**
- **Named the policy as EC2-Region-Restricted.**
- **Created the policy successfully.**



- **Attached the policy to the IAM user.**

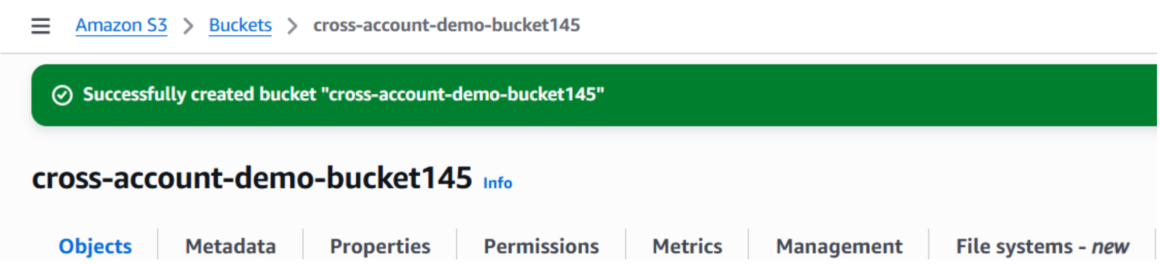


- **Verified that EC2 access is allowed only in the test-user.**



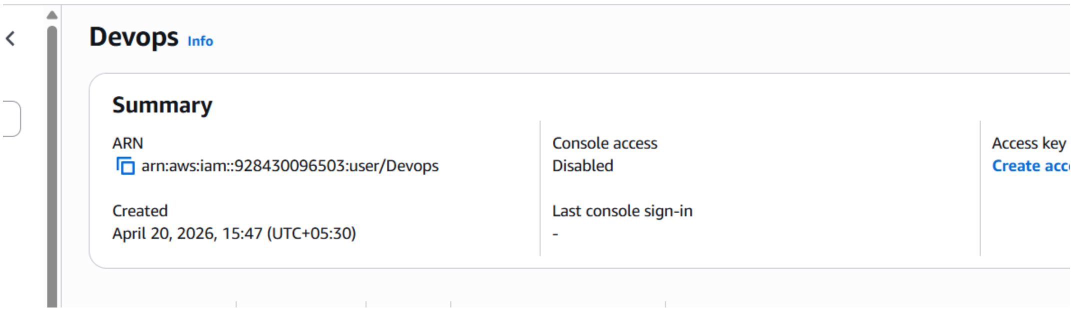
6. We have two accounts: Account A and Account B. Account A user should access an S3 bucket in Account B.
(Collaborate with a team member and execute this. This is mostly asked in every interview.)

- **Collaborated with a team member using two AWS accounts.**



- Created an S3 bucket in Account B.

- Configured bucket policy to allow access to IAM user from Account A.



- Defined Principal using IAM user ARN.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowListBucket",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::009365819224:user/test-user"
      },
      "Action": "s3:ListBucket",
      "Resource": "arn:aws:s3:::cross-account-demo-bucket145"
    },
    {
      "Sid": "AllowObjectAccess",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::009365819224:user/test-user"
      },
      "Action": [
        "s3:GetObject",
```

- Granted required S3 permissions (ListBucket, GetObject)
- Attached IAM permissions to user in Account A.

Policy editor

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "ListBucket",
            "Effect": "Allow",
            "Action": "s3:ListBucket",
            "Resource": "arn:aws:s3:::cross-account-demo-bucket0408"
        },
        {
            "Sid": "GetObject",
            "Effect": "Allow",
            "Action": "s3:GetObject",
            "Resource": "arn:aws:s3:::cross-account-demo-bucket0408/*"
        }
    ]
}
```

- **Verified access using AWS CLI (aws s3 ls).**

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ aws configure
AWS Access Key ID [****************FDNU]: AKIA5QKV7AB3XV2AZJWD
AWS Secret Access Key [****************OsXw]: 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
Default region name [us-east-1]: us-east-1
Default output format [json]: json

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ aws sts get-caller-identity
{
    "UserId": "AIDA5QKV7AB3TBVIT2KUT",
    "Account": "928430096503",
    "Arn": "arn:aws:iam::928430096503:user/Devops"
}

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ aws s3 ls s3://cross-account-demo-bucket0408
2026-04-20 18:36:54        293 create_user.bash

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$
```

- Successfully accessed cross-account S3 resources.